

Absolute Stability

$$u'(t) = \lambda u$$

$$u(0) = \eta$$

$$u(t) = e^{t\lambda} \eta$$

For Euler's method:

$$E^{n+1} = (1+k\lambda)E^n - k\tau^n$$

$$\Rightarrow \|E^N\| \leq T e^{T|\lambda|} \|\tau\|_{\infty}$$

Notice: The global will grow at each step $|1+k\lambda| > 1$.

$$u'(t) = -\sin(t)$$

$$u(0) = 1$$

$$u(t) = \cos(t)$$

$$L = 0$$

Prothero-Robinson

$$u'(t) = \lambda(u - \cos(t)) - \sin(t)$$

$$u(0) = 1$$

$$L = |\lambda|$$

$$u(t) = \cos(t)$$

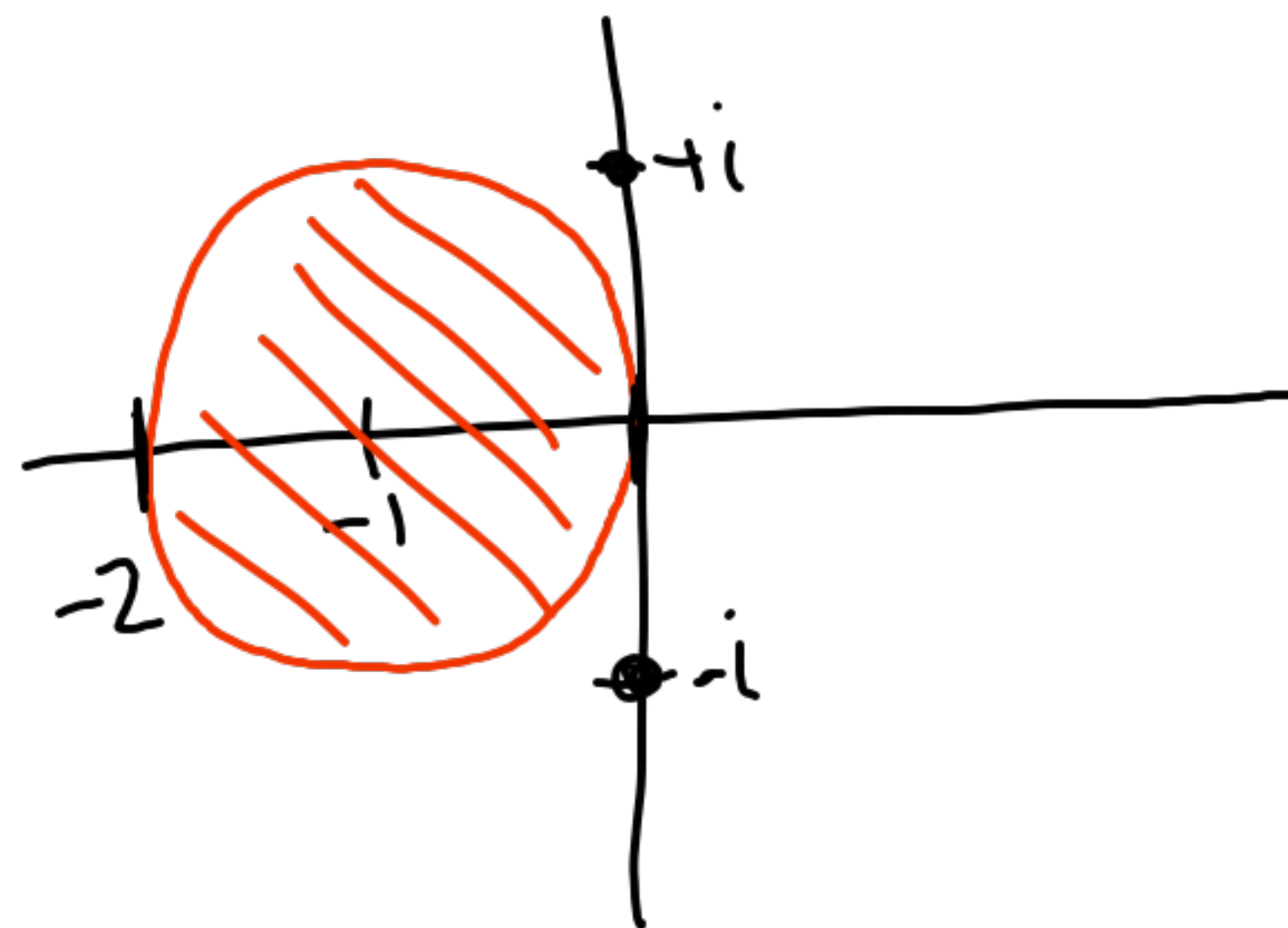
$$|1 + k\lambda| \leq 1$$

$$-1 \leq 1 + k\lambda \leq 1$$

$$\frac{-2}{\lambda} \geq k \geq 0 \quad (\lambda < 0)$$

$$\frac{-2}{-250} = \frac{1}{125}$$

$$z = k\lambda$$



$$S = \{z \in \mathbb{C} : |z - (-1)| \leq 1\}$$

Linearized Pendulum

$$\Theta''(t) = -\sin(\Theta(t)) \approx -\Theta(t)$$

$$\Theta' = v(t) \quad v' = \Theta'' = -\Theta$$

$$u = \begin{bmatrix} \Theta \\ v \end{bmatrix}' = \underbrace{\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}}_A \begin{bmatrix} \Theta \\ v \end{bmatrix}$$

$$\begin{bmatrix} u(0) \\ v(0) \end{bmatrix} = \eta$$

$$\begin{aligned} \lambda^2 + 1 &= 0 \\ \lambda &= \pm i \end{aligned}$$

$$\text{Solution: } u(t) = e^{tA} \eta$$

$$A = R \Lambda R^{-1} \quad \Lambda = \begin{bmatrix} -i & 0 \\ 0 & +i \end{bmatrix}$$

$$e^{tA} = R e^{t\Lambda} R^{-1}$$

For the problem $u'(t) = Au(t)$

($A = \mathbb{R}^{n \times n}$), Euler's method

gives $E^{n+1} = (I + kA)E^n - k\tau^n$

So the error will grow if

$$\|I + kA\| \geq 1$$

Taking the 2-norm,
We need the eigenvalues
of $I+kA$ to be < 1
in magnitude.

If λ is an e.v. of A ,
the e.v.'s of $I+kA$ are
 $1+k\lambda$, so we need
 $|1+k\lambda| \leq 1$.

Implicit Euler

$$U^{n+1} = U^n + k f(U^{n+1})$$

$$f(u) = \lambda u$$

$$U^{n+1} = U^n + k\lambda U^{n+1}$$

$$(1 - k\lambda)U^{n+1} = U^n$$

$$U^{n+1} = \frac{1}{1 - k\lambda} U^n$$

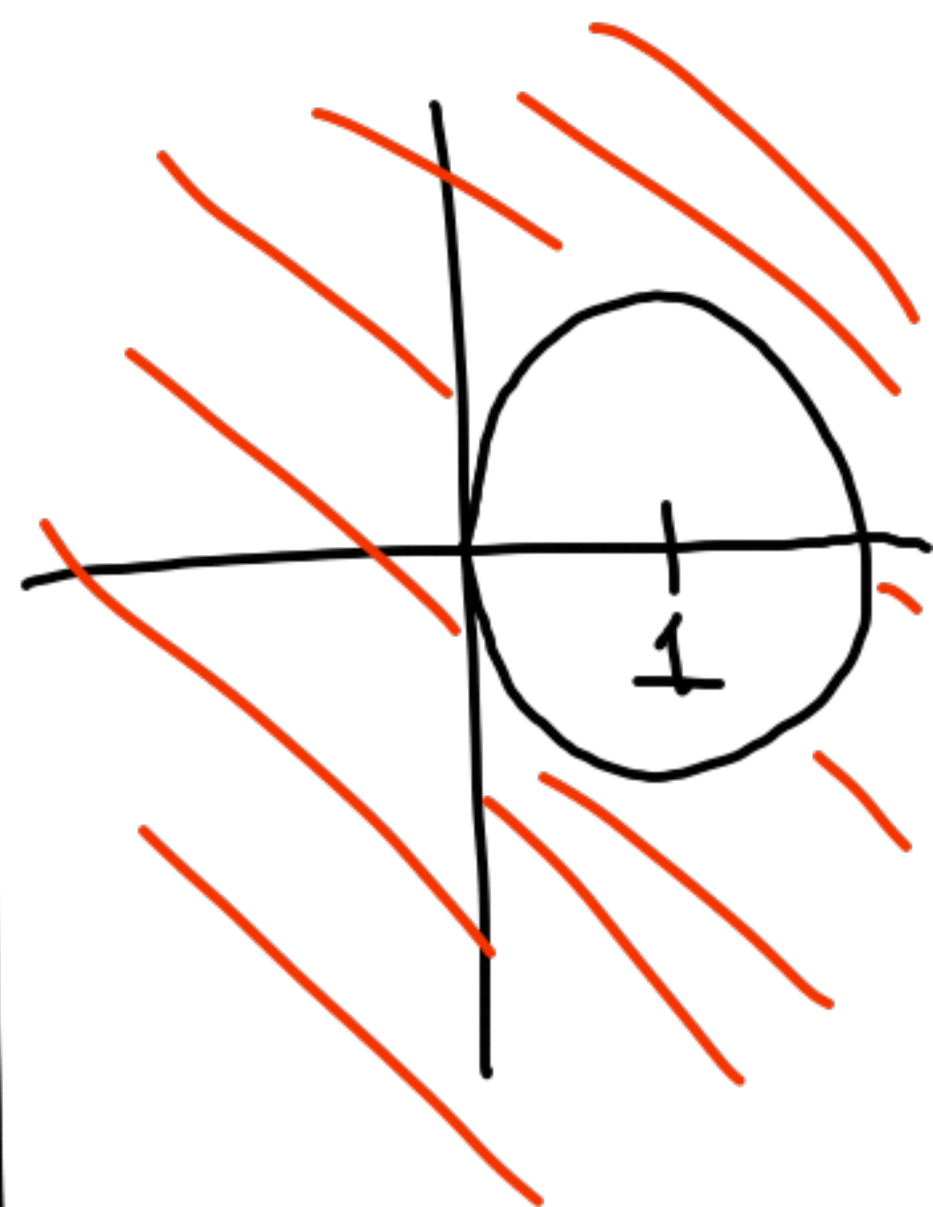
The global error satisfies

$$E^{n+1} = \frac{1}{1 - k\lambda} E^n - k\tau^n$$

We need

$$\left| \frac{1}{1 - k\lambda} \right| \leq 1$$

$$1 \leq |1 - z|$$



Any one-step method
applied to

$$u'(t) = \lambda u$$

yields

$$U^{n+1} = R(k\lambda) U^n$$

$$E^{n+1} = R(k\lambda) E^n - k \tau^n$$

where $R(z)$ is stability
function of the method.

Exp. Euler: $R(z) = 1 + z$

Imp. Euler: $R(z) = \frac{1}{1-z}$

Region of absolute stability:

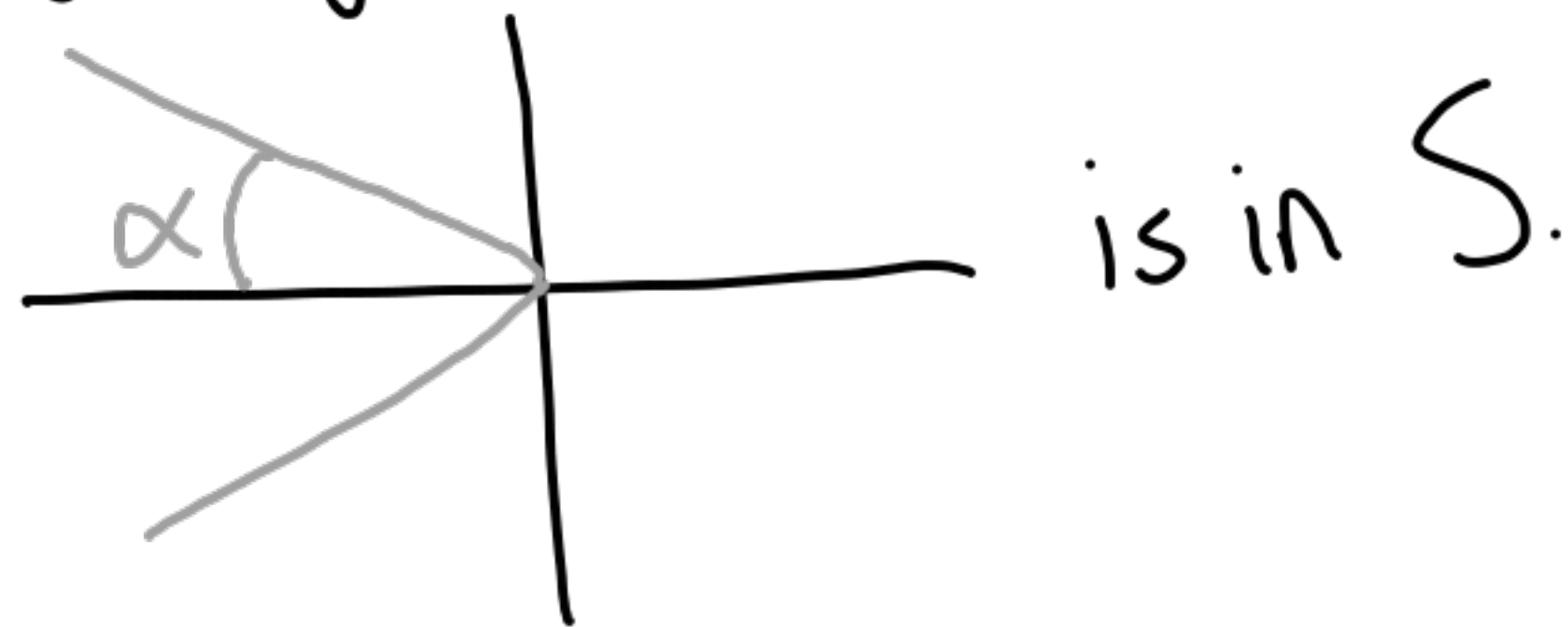
$$S = \{z \in \mathbb{C} : |R(z)| \leq 1\}$$

If we choose k s.t.

$|R(k\lambda)| \leq 1$, we say the
method is absolutely stable.

If $\mathbb{C}^-$ (the closed left half plane) is in S , we say the method is A -stable.

We say a method is $A(\alpha)$ -stable if the wedge (sector)



Multistep Methods

$$\sum_{j=0}^r \alpha_j U^{n+j} = k \sum_{j=0}^r \beta_j f(U^{n+j})$$

$$f(u) = \lambda u:$$

$$\sum_{j=0}^r (\alpha_j - k\lambda \beta_j) U^{n+j} = 0$$

$$U^n = \rho^n: \sum_{j=0}^r (\alpha_j - z \beta_j) \rho^j = 0$$

$$z = k\lambda$$

$\pi(\xi; z)$: 2nd char. poly.

The global error satisfies

$$\sum_{j=0}^r (\alpha_j - z \beta_j) E^{nj} = k \tau^n$$

So we need $\pi(\zeta, z)$ to satisfy the root condition:

① $|\zeta| \leq 1$

② If $|\zeta| = 1$, ζ is a simple root.

$$S = \left\{ z \in \mathbb{C} : \pi(\zeta, z) \text{ satisfies the root condition} \right\}$$

On the boundary of S , we have $|\zeta| = 1$ for some root.

Set $\zeta = e^{i\theta}$:

$$\pi = \sum_{j=0}^r (\alpha_j - z \beta_j) e^{i\theta j} = 0$$

$$\sum_{j=0}^r \alpha_j e^{i\theta j} = z \sum_{m=0}^r \beta_m e^{i\theta m}$$

$$Z = \frac{\sum_j \alpha_j e^{i\theta_j}}{\sum_m \beta_m e^{i\theta_m}} \left. \begin{array}{l} \text{Evaluate this} \\ 0 \leq \theta \leq 2\pi \end{array} \right\}$$

This gives the boundary of S .